



**AMITY INSTITUTE OF
INFORMATION TECHNOLOGY
(AIIT)**

Project Report

on

**Employee Burnout Prediction
Using Machine Learning**

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1. Abstract

Employee burnout has become a critical challenge for organizations due to increasing workloads, workplace stress, and changing work environments. Burnout negatively affects employee productivity, job satisfaction, mental health, and overall organizational performance. Early identification of burnout enables organizations to take preventive measures before employees experience severe stress or reduced performance.

The objective of this project is to develop a web-based Employee Burnout Prediction System using Machine Learning. The system predicts employee burnout levels based on work-related and personal factors such as gender, company type, work-from-home availability, designation level, resource allocation, and mental fatigue score. A Random Forest Regressor algorithm is trained using historical employee data to estimate burnout scores accurately.

The application is developed using Python and Flask for backend processing, while HTML, CSS, and JavaScript provide a responsive and user-friendly interface. The system supports both individual employee prediction and bulk prediction through CSV file uploads. It also provides burnout classification, personalized recommendations, and feature importance analysis to improve transparency and usability.

The developed system demonstrates how Machine Learning can be effectively integrated with Human Resource Analytics to automate burnout assessment. It provides a practical decision-support solution that assists HR professionals in identifying employees at risk and improving overall workforce well-being.

2. Introduction

Employee burnout is a psychological condition resulting from prolonged workplace stress, excessive workload, lack of motivation, and poor work-life balance. It has become one of the major concerns for organizations because it reduces employee productivity, increases absenteeism, and contributes to higher employee turnover. Traditional methods of evaluating

burnout, such as employee surveys and managerial observations, are often subjective, time-consuming, and difficult to scale in large organizations.

Recent developments in Artificial Intelligence and Machine Learning have enabled organizations to analyze employee data more effectively and identify burnout patterns with greater accuracy. Machine learning models can process multiple employee-related factors simultaneously and detect hidden relationships that are difficult to identify manually. This allows organizations to take preventive actions before burnout significantly impacts employee performance.

This project presents a web-based Employee Burnout Prediction System that uses a Random Forest Regressor to estimate employee burnout scores. Users can enter employee information manually or upload multiple employee records using CSV files. The system predicts burnout levels, categorizes employees into Low, Medium, or High risk, and provides suitable recommendations.

The project demonstrates the practical application of Machine Learning in Human Resource Analytics while providing an efficient, scalable, and user-friendly solution for burnout prediction. It also highlights the potential of AI-based decision-support systems in improving employee well-being and organizational productivity.

3. Methodology

The Employee Burnout Prediction System follows a systematic machine learning workflow consisting of data collection, preprocessing, model training, prediction, and result visualization.

Initially, the employee dataset is loaded and validated. Data preprocessing includes selecting relevant features, encoding categorical variables using Label Encoding, and standardizing numerical features through StandardScaler. These preprocessing steps ensure that the dataset is suitable for machine learning algorithms.

The processed dataset is divided into training (80%) and testing (20%) subsets. A Random Forest Regressor is then trained using the training dataset to predict continuous burnout scores. The trained model is

evaluated using regression performance metrics such as R^2 Score, RMSE, and MAE. After successful training, the model is stored using Joblib, allowing it to be loaded directly during application startup without retraining.

The web application is developed using the Flask framework. Employee details entered through the user interface or uploaded as CSV files are preprocessed using the same transformation pipeline before being passed to the trained model. The predicted burnout score is classified into Low, Medium, or High burnout levels based on predefined thresholds. The application also calculates feature importance to help users understand which employee attributes contribute most significantly to the prediction.

The overall architecture consists of a Presentation Layer (HTML, CSS, JavaScript), an Application Layer (Flask), and a Machine Learning Layer (Random Forest Model), ensuring modularity and efficient system performance.

4. Implementation and Results

The Employee Burnout Prediction System is implemented as a responsive web application using Python, Flask, HTML, CSS, and JavaScript. The application consists of four primary modules: Home Page, Single Employee Prediction, Bulk CSV Prediction, and Model Information Dashboard.

The Single Prediction module allows HR personnel to enter employee details manually and receive an instant burnout prediction along with personalized recommendations. The Bulk Prediction module enables organizations to upload CSV files containing multiple employee records, allowing simultaneous prediction and analysis of burnout levels across large workforces.

The backend performs input validation, data preprocessing, feature transformation, prediction, and recommendation generation. The trained Random Forest Regressor predicts a continuous burnout score, which is categorized into Low, Medium, or High burnout risk. The application also displays feature importance analysis, highlighting the influence of variables such as Mental Fatigue Score, Resource Allocation, and Designation Level on prediction outcomes.

The system achieved strong prediction performance and demonstrated good generalization on unseen data. Compared with traditional manual assessment methods, the application provides faster, more consistent, and data-driven results. Interactive dashboards, workflow diagrams, and prediction interfaces further enhance usability and improve transparency for end users.

5. Conclusion

This project successfully demonstrates the application of Machine Learning in predicting employee burnout using a Random Forest Regressor. The developed web application automates burnout assessment by analyzing employee-related factors and generating accurate burnout predictions with personalized recommendations.

The integration of Machine Learning with Flask provides an efficient and user-friendly platform suitable for both educational purposes and practical Human Resource Analytics. The system reduces manual effort, improves decision-making, and helps organizations identify employees who may require timely support.

Although the current implementation provides reliable predictions, its accuracy depends on the quality of the training dataset and the selected input features. Future enhancements may include integration with HR Management Systems, Explainable AI techniques such as SHAP, cloud deployment, user authentication, mobile applications, and continuous model retraining using real-time organizational data.

Overall, the Employee Burnout Prediction System provides a scalable, practical, and intelligent solution that demonstrates the potential of Artificial Intelligence in improving employee well-being and organizational productivity.

6. References

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